INCH-POUND
MS17243E
13 June 2011
SUPERSEDING
MS17243D
18 November 2004

DETAIL SPECIFICATION SHEET

VALVES, ANGLE, SHUT-OFF, PACKED, RECEIVER, REFRIGERANT-12

Inactive for new design after 6 February 1996.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described shall consist of this specification.

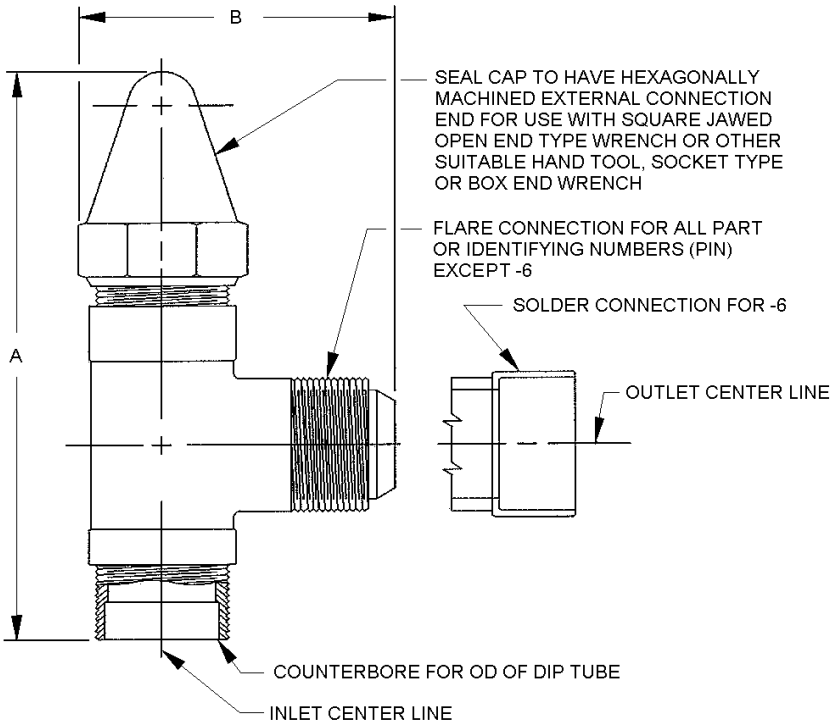


FIGURE 1. Angle shut-off receiver valve, refrigerant-12.

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TABLE I. Valve dimensions. 1/ 2/

Dash number	Size inlet NPT	Size outlet		A max (mm)	B max (mm)
		OD flare	OD solder		
-1	1/4	1/4	-	3.625 (92.08)	1.625 (41.28)
-2	3/8	3/8	-	4.000 (101.60)	2.000 (50.80)
-3	3/8	1/2	-	4.2509 (107.98)	2.000 (50.80)
-4	1/2	1/2	-	4.500 (114.30)	2.250 (57.15)
-5	1/2	5/8	-	5.000 (127.00)	2.500 (63.50)
-6	3/4	-	7/8	6.000 (152.40)	2.750 (69.85)
-7	1/2	3/8	-	4.500 (114.30)	2.250 (57.15)

1/ Dimensions are in inches.
2/ Metrics are given for general information only.

REQUIREMENTS

Intended use. Valves covered in this specification are intended for use in a refrigerating system as a means of shutting off the system's charge of refrigerant and containing it in the liquid receiver or condenser receiver.

Materials. Valve body and seal cap - brass or steel.
Finish – brass – none, steel – zinc plated.
Ring gasket - soft copper.
Valve stem (including integral plug) - commercial bronze or steel treated to resist corrosion.

Operating conditions. Maximum operating pressure: 350 psig (2.41MPa).
Maximum operating temperature: 200°F (93.3°C).

Connection ends. Inlet - Male pipe thread in accordance with SAE-J513.
Outlet (all except -6) - Flare tube male thread in accordance with SAE-J513.
Outlet (-6 only) - Solder tube female socket in accordance with ASME B16.22 for OD tube size 7/8 inch.

Type. Angle, packed, back-seating or non-back-seating key-operated plug element, with gasketed seal cap, valve stem shank to be 3/16, 1/4, or 5/16 inch square shape.

Identification of product. The PIN for the valve consists of the MS number plus the dash number.

Example: MS17243-1-B.
- B — Material, finish (see table II)
- 1 — Size (see table I)
- MS17243 — Drawing number

Standard product. Valves delivered under this specification shall be the manufacturer's standard commercial catalog product, except for any modifications and additional requirements specified herein.

Design and construction. The valves shall be of the angle, packed, back-seating, or nonbackseating type with a gasketed seal cap. A wrench grip shall be provided on each valve. Threads and the seating surface of the flare connection shall be protected with plastic caps or by other means to prevent damage during handling.

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Material. Material shall be brass or steel.

TABLE II. Material, finish.

Dash Number	Material	Finish
B	Brass	none
C	Steel	ASTM B633, type 6

Marking. The manufacturer's standard identification and other data standard with the manufacturer shall be placed on all valves. Unmarked valves will be accepted where the manufacturer's standard practice does not require marking.

Workmanship. All parts of the finished valves shall be clean and free from pits, sprues, scale, flux or extraneous material. External surfaces shall be free from burrs and sharp edges. Passages shall be clean and free from pits, sprues, scale, flux and foreign material.

Angle valve leak test. The angle valve, with the seat open and the discharge end blocked, shall be submerged in water and a pressure of 350 pound-per-square-inch gage pressure of air shall be applied to the intake to check for body leakage. The pressure shall be released and the seat closed and the discharge end unblocked. The valve shall be submerged in water again and the same pressure applied. Any leakage through the body or by the seat shall be cause for failure of this test.

Order of precedence. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

Note. The procurement specification, MIL-V-22854, was cancelled without replacement on 16 February 1993.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. This document references the following:

ASME B16.22
ASTM B633
SAE-J513

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CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – SH
Air Force – 99
DLA – CC

Preparing activity:
DLA - CC

(Project 4820-2011-004)

Review activities:
Army – AT, CR4
Navy – MC, SA, YD
Air Force - 71

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